

# Inside the Channel: A Stochastic Analysis of the CSI for Wi-Fi Sensing

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**Abstract**—With the widespread interest in Joint Communication and Sensing, a key point of new generations of wireless networks, comes the need for a thorough analysis of the information that sensing tools leverage. Although multiple works focus on the use of Wi-Fi Channel State Information (CSI) to implement ambient sensing techniques and tools, a statistical characterization of CSI structure and behavior across variable environmental conditions is still missing. This work analyzes the fundamental properties of the CSI through a focus on the long-term behavior of its amplitude and its interpretation as a stochastic process. We show that the short-term amplitude variations, the increments, are accurately modeled by a heavy-tailed T Location Scale distribution across a wide range of real-world experimental conditions, regardless of bandwidth, hardware, and environmental dynamics. The analysis of amplitude autocorrelation further reveals the presence of residual memory beyond the mean value in the amplitude process, indicating a non-Markovian evolution. The increments autocorrelation, which exhibits one-step memory, is analytically consistent with this finding. Building on this stochastic characterization, we introduce a formal framework to uniform CSI representation and we define a simple distance-based metric to support the comparison of CSI samples collected under heterogeneous experimental conditions. The metric is then used by a proof-of-concept lightweight sensing method presented in the final part of the study: amenable to implementation on resource-constrained devices typical of the Internet of Things, the algorithm is able to accurately discriminate static from dynamic environments showing how the stochastic characterization of CSI's can inform training-free passive CSI-based sensing with Wi-Fi without requiring complex learning models.

**Index Terms**—CSI Sensing; Wi-Fi; Joint Communication and Sensing; Stochastic Analysis; Open-source dataset

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## I. INTRODUCTION

IN recent years, the concept of Integrated or Joint Communication and Sensing (JCAS) has definitely entered the perimeter of Internet of Things (IoT) research, representing one further step toward a pervasive presence of the Internet in everyday life. Since early research on the subject, as [1], [2], [3], [4], JCAS has evolved to encompass many different flavors of exploiting the same portion of the electromagnetic spectrum to simultaneously transmit information and measure some properties of the environment. The interesting survey presented in [5] identifies ten open challenges in JCAS, ranging from theoretical limits to system design and privacy protection. It offers a comprehensive discussion of the many facets of JCAS, and a contextualization of research in this field.

We agree with all the identified challenges, although it is possible that additional ones exist; however, it must also be noted that different interpretations of the JCAS domain may lead to entirely distinct approaches and results. For instance, the theoretical limits vary significantly – and possibly require different approaches – depending on whether JCAS refers to measurements involving a collaborating User Equipment (UE) or to measurements of the environment or people who do not carry any device. Similarly, privacy concerns are entirely different, and, as highlighted in [6], the corresponding countermeasures require radically different methodologies.

The access technology too, whether circuit- (as in cellular networks) or packet-oriented (as in wireless Local Area Networks (LANs)), influences both the methodologies and the reachable limits. Furthermore, the decision to either use exactly the same signals for sensing and information transfer or to exploit different, possibly coordinated signals at different times for the two purposes – as explored in [7] – changes the design and analysis perspective completely.

This work focuses on sensing functions with standard Wi-Fi: we target a packet-based technology in which sensing exploits the same signals used for communication. More specifically, sensing is performed by leveraging the Channel State Information (CSI) derived from the frame preambles, which are typically used for channel equalization before decoding. It should be noted that, since 3GPP technologies use Orthogonal Frequency Division Multiplexing (OFDM) modulations as well, extending the analysis to 5G and beyond is feasible, if not straightforward.

Most works in the area of CSI-based JCAS use supervised and/or unsupervised Artificial Intelligence (AI) techniques to extract the information from the CSI. The approaches range

from Deep Neural Networks (DNNs) [8] to auto-encoders [9], [10], to Doppler analysis for motion recognition [11], [12], [13], and many more. Attention has recently been paid to the potential that larger bandwidths in new Wi-Fi standards can bring to sensing: IEEE 802.11ax [14] introduced the subcarrier spacing of 78.125 kHz and supports channel bandwidths up to 160 MHz, while 802.11be [15] introduced 320-MHz channels [16]; furthermore, IEEE recently published the CSI-based sensing standard version, 802.11bf [17].

All this research is leading to impressive advances and promising innovations; however, we deem that a full understanding of the potential of CSI for JCAS is hindered by the lack of analysis and models of CSI properties beyond channel coherence. The recent analysis in [18] supports and informs this statement, presenting results that show the difficulty of generalizing sensing results and extending them in time.

This paper – which extends and completes our presentation in [19] – is a first endeavor to address the issue of empirically quantifying the amount of information embedded in CSI samples for sensing purposes.

Compared with [19], this work introduces five new, key contributions:

- A theoretical and empirical analysis of the CSI amplitude increments showing that their distribution is heavy-tailed and therefore deviates from the behavior predicted by classical Rayleigh/Rician amplitude models. We interpret this mismatch as the effect of time-varying channel properties and model the increments with a T Location Scale distribution.
- An analytical characterization of the relationship between CSI amplitude and increment autocorrelation, highlighting the presence of memory and suggesting a non-Markovian amplitude evolution.
- An extensive validation of the proposed stochastic model over 70 experiments, collected over several months across different bandwidths, frame rates, and antenna configurations. The model is further validated on a third-party dataset, comprising CSI's collected in multiple environments with different hardware and experimental setups.
- The design of an interpretable, lightweight ambient sensing algorithm for occupancy detection, derived from the proposed stochastic framework, together with its validation on both datasets.
- The comparison of the proposed algorithm against a state-of-the-art Neural Network (NN)-based method on a public dataset, and against an elementary Support Vector Machine (SVM) algorithm on the dataset we collected, including performance metrics, the analysis of computational complexity, and execution times.

The key takeaway is the empirical evidence that the stochastic properties of the CSI enable simple, computationally efficient sensing algorithms to reliably discriminate among real-world scenarios beyond lab-tuned human activity recognition. We remark that the contribution of this work is not the proposal of yet another CSI-based sensing or classification algorithm, but the basic stochastic analysis of the CSI properties themselves, which can complement and inform sophisticated sensing and classification algorithms.

For the sake of clarity, we will always use the term *scenario* to identify a situation that we wish to be able to sense and isolate from any other: different rooms are different scenarios, as well as an empty room is a different scenario from the same room with any number of people inside. For instance, if the considered application is localization, each position of a person would be a specific *scenario* to be distinguished from the other positions. We instead call *experiment* a single session of measurement in a scenario. For example, in a localization application using Machine Learning (ML), the training and testing datasets contain different experiments over the same set of scenarios.

The goal of this work is to provide a stochastic characterization of the channel through modelling of CSI statistics. Thus, the analysis is intentionally focused on accurately describing the empirical behavior of the CSI rather than deriving a predictive physical channel model. In this context, the descriptive nature of the proposed stochastic model is a deliberate design choice whose goal is to capture statistically relevant CSI features that can be leveraged by sensing applications.

## II. RELATED WORK

CSI sensing can deal with a wide range of applications, from localization to human detection, to gesture and activity recognition. Current investigations usually employ Deep Learning (DL) methods to target one specific application at a time [20]. However, even though the results are often impressive in terms of accuracy, there are few (if any) efforts to analyze the limitations of these approaches. In general, DL techniques rely on the collection of very large, labeled datasets for each target application, which is a labor-intensive, time-consuming endeavor. In this case, while cross-domain Wi-Fi sensing can mitigate difficulties in data collection activities [21], it remains unclear whether it provides reliable indications on how sensing algorithms perform across different scenarios. As our focus is not on DL methods, the interested reader can find exhaustive surveys in the aforementioned [5], [20] as well as [22] and [23].

Traditionally, RF-based sensing and localization techniques have also been using observation variables that are much coarser than the CSI. For instance, the Received Signal Strength Indicator (RSSI) is a measure that is often simpler to acquire, but it collapses the frequency-selective behavior of the channel into one single scalar, whereas the CSI provides a “richer” per-subcarrier information. Recent works continue exploring this broader area, either by exploiting generic RF characteristics with auxiliary infrastructure [24] or by surveying RSSI-based fingerprinting approaches for indoor positioning [25]. However, while RSSI remains an important reference point for very-low-cost deployments, CSI can be practically obtained even in low-power IoT scenarios, thus making the analysis more sophisticated and appealing at no additional cost.

Here we focus on works that address CSI sensing through either interpretation of physical models or algebraic and probabilistic methods. In the first group, we essentially find papers that exploit Doppler effects. The authors of [26] introduce the

notion of differential CSI (named TD-CSI or Time-domain Difference of CSI) that amplifies the CSI components due to variations in the channel propagation paths. Similarly, [11] focuses the comparison of activity recognition through AI-based Doppler analysis. Trajectory estimation through Doppler frequency analysis is the focus of [27], [28]. To improve Doppler estimation of propagation components due to a moving object (typically a person), the authors propose using multiple measurement points, called *responders*, that exploit a request/response protocol initiated by an *initiator* to cancel phase offsets that hamper Doppler computation. A different way to exploit Doppler effects is presented in [12], where they are used to remove the CSI dependence on the specific ambient (room, furniture, etc.) and then feed the AI model, which is trained to recognize human activities.

In the second group, to which this paper and our presentation [19] belong, we find articles that do not exploit physical properties, but rather represent the CSI's using interpretable models that also lend themselves to what-if analysis. The approach in [29], [30] exploits the stochastic analysis of the electromagnetic field, modifying the CSI's under the assumption that human bodies are moving reflectors. Other authors focus on the autocorrelation function of the CSI samples. The goal of [29] is intrusion detection, while the goal of [30] is to distinguish among activities characterized by movements at various speeds. A different approach is presented in [31], which proposes to discern an empty area from an occupied one through the temporal variations of CSI samples; the key focus is on the CSI collection (or sampling) process to reduce noise and outliers by using sequential Monte Carlo sampling.

In state-of-the-art works, the representation of the stochastic properties of the CSI is embedded inside the DL model [32]; however, such a representation has two major shortcomings: first, it is not general by definition, as it is strongly tied to the specific DL model and not to the statistical properties of the CSI; second, and more important, the DL-based representation is extremely complex, and thus it cannot be easily analyzed nor explained. The authors of [33] explicitly identify how CSI features vary with environmental dynamics – showing that they become more variable in more dynamic settings – but then rely on ML techniques to perform activity recognition.

Traditional statistical methods are used in [34] to select those CSI subcarriers that contribute the most to the accuracy of a breath detection application. The analysis aims to improve existing DL algorithms and does not explore other questions.

To close our review, we mention [35], [36]. The first studies the quality of CSI extraction, proposing a method to make the CSI representation independent from the transmission mode; the authors show that applying this extraction procedure to existing classification models [37] improves performance. The second focuses on the impact of the CSI collection rate on the accuracy of the results for motion detection applications. Both papers highlight how, despite the large amount of works on JCAS, there is still a need to carry out foundational studies to propose interpretative models to drive adoption in engineering solutions.

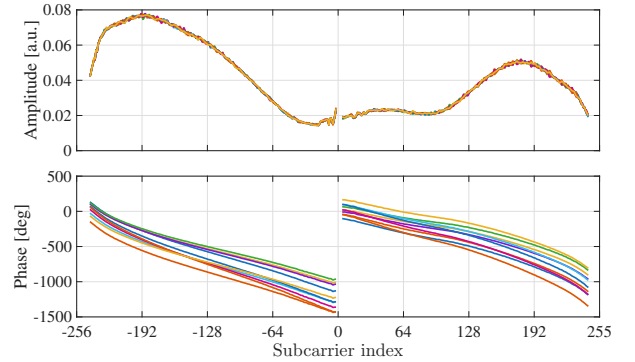


Figure 1. Amplitude (top) and unwrapped phase (bottom) of 10 different CSI samples collected during a 10-minute experiment (code E40.0/0, see Tab. II). CSI's are normalized by the integral of the energy to remove the effect of the AGC. Subcarriers suppressed in transmission are discarded.

### III. CSI ESTIMATION AND SENSING BACKGROUND

Each experimental scenario considered in this work consists of collecting  $M$  CSI of Wi-Fi frames transmitted over  $N_{sc}$  OFDM subcarriers, where

$$k \in [1, M], \quad n \in \left[ -\frac{N_{sc}}{2}, \frac{N_{sc}}{2} - 1 \right]$$

are respectively the (time) index of the Wi-Fi frame and the OFDM subcarrier index. Thus, the CSI  $\mathbf{C}$  is the representation of the channel distortion measured at the receiver:

$$\mathbf{C}(k, n) = |\mathbf{C}(k, n)| \cdot e^{j\angle \mathbf{C}(k, n)}.$$

Figure 1 shows the amplitude  $|\mathbf{C}|$  (top) and phase  $\angle \mathbf{C}$  (bottom) of 10 different CSI samples collected at random times during experiment E40.0/0 (see Tab. II). The CSI's are remarkably similar to one another, and this similarity, together with subtler features not always easy to identify, is what DL algorithms exploit to classify different scenarios.

Albeit usually described as the “channel response”, the CSI is a local estimation, normally extracted with some tool from the receiver chipset and represented in an arbitrary measure unit ([a.u.]) as IQ components of the received signal. The original extraction is based on a fixed point representation of the Analog-to-Digital Converter (ADC) component of the chipset that is unfortunately unknown to the extractor.<sup>1</sup> The extracted values, affected by changes in the Automatic Gain Control (AGC) that cannot be controlled, are cast onto floating point numbers. The goal of *sensing* is usually the recognition of the same scenario in *collections* of CSI samples measured at different moments. Typically,  $|\mathbf{C}|$  (alone, or with the phase  $\angle \mathbf{C}$ ) or the IQ values are fed to DL algorithms for training and testing purposes. The number of CSI samples  $M$  is normally in the order of some thousands for training purposes and as small as possible for testing purposes (but still compatible with achieving the desired accuracy).

Model-based scenario identification seems unfeasible at present; however, this is no reason to stop investigating

<sup>1</sup>We use an extension of the Nexmon CSI Extractor tool, which was presented for the first time in [38] and later extended in collaboration with the SEEMOO research group at the University of Darmstadt.

whether CSI's can be analyzed and interpreted through an appropriate lens that gives insight into the sensing process, informs, and directs it.

From the perspective of a generic observer,  $\mathbf{C}(k, \cdot)$  with  $k \in [1, M]$  is a sample of a multi-varied random process with unknown distributions, dependent on the specific scenario  $S_j$  considered. The process is composed by  $2N_{sc}$  variables: amplitude and phase of each subcarrier. More appropriately, a collection of CSI samples  $\mathbf{C}$  is a realization of a Discrete-Time Discrete-Space (DTDS) random process sampled from the specific subset of the CSI population conditioned on a particular scenario  $S_j$ . The marginal distributions and the correlation structure of this process are additional pieces of information that DL algorithms exploit to sense the environment, i.e., to distinguish the realizations of this process that are conditioned on different scenarios.

For simplicity, in this paper we limit the analysis to the amplitude  $|\mathbf{C}(k, \cdot)|$ , which we rename  $A(k, \cdot)$ , leaving the analysis of the phase for future work. This is a deliberate simplification, not a claim that the phase is uninformative. Indeed, while phase can be highly valuable for sensing and localization [4], [39], commodity Wi-Fi phase measurements are also markedly more sensitive to synchronization offsets, packet-detection jitter, and extractor-dependent calibration [40]. However, since the objective of this work is a long-term stochastic characterization, our focus is only on the amplitude, which can be normalized more robustly in the considered datasets.

Of particular interest for the rest of the paper are the process of the amplitude increments on the single subcarrier  $n$ :

$$\delta(t, n) = A(t, n) - A(t - 1, n) \quad (1)$$

with  $t \in [2, M]$ , and the auto-covariance of the amplitude on the single subcarrier  $n$ :

$$R_A(\tau, n) = \mathbb{E}[A(k, n) \cdot A(k - \tau, n)] - \mu_{A(\cdot, n)}^2 \quad (2)$$

where  $\tau \in [0, \lfloor M/2 \rfloor]$ , assuming the DTDS of the CSI is locally (limited to the single experiment) wide-sense stationary. In fact, if this assumption does not hold, the possibility of ‘sensing’ exploiting the CSI properties fails. From the normalization of the auto-covariance (Eq. (2)), we derive the equation for the autocorrelation of the amplitudes of the single subcarrier  $n$ :

$$\rho_A(\tau, n) = \frac{R_A(\tau, n)}{R_A(0, n)} = \frac{R_A(\tau, n)}{\sigma^2} \quad (3)$$

In this analysis,  $\tau$  is limited to  $\lfloor M/2 \rfloor$  to have a good estimate of  $R_A$  and  $\rho_A$ . We indicate with  $\mu_{A(\cdot, n)}$  the sample mean of the  $n$ -th subcarrier. The same computations can be applied to  $\delta$  to obtain the auto-covariance  $R_\delta$  and autocorrelation  $\rho_\delta$  of the subcarriers increments.

Given the CSI extraction procedure and the differences that arise from distinct bandwidths, technologies, and devices, the CSI must be properly normalized to be able to make meaningful comparisons. The detailed description of the process can be found in [41], but in brief, it entails the following three steps: *i)* Normalize the energy of each CSI to compensate for the AGC variations; *ii)* Find the maximum and the minimum of  $A(\cdot, \cdot)$  in each experiment; *iii)* Map the amplitudes of all the

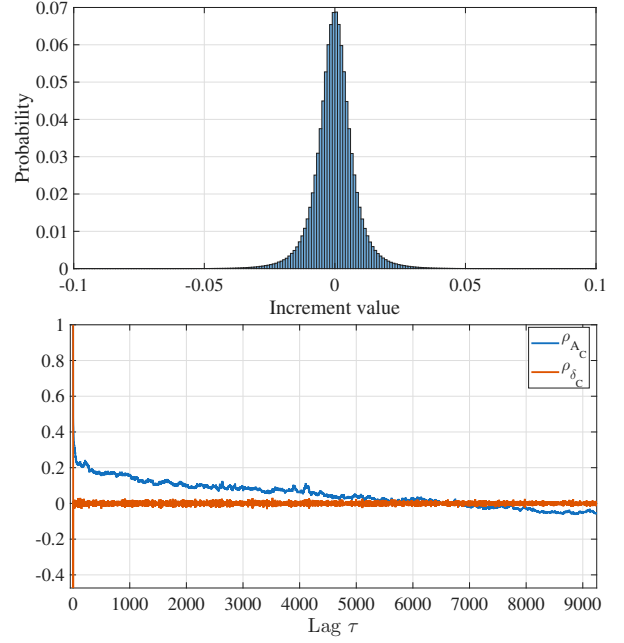


Figure 2. Basic properties of subcarrier amplitudes (experiment E40.0/0). Top: Distribution of the increments (Eq. (1)). Bottom: Autocorrelation of the amplitude (blue) and the increments processes (red) according to Eq. (3).

CSI in the experiment in the interval  $[0, 1]$ . To avoid cluttering the notation, from now on  $A$  and  $\delta$  refer to the output of this normalization process.

Figure 2 reports some basic properties of a single subcarrier for a typical experiment performed on a 40 MHz-bandwidth channel after  $A$  has been normalized as described above. The top plot reports the distribution of the increments (Eq. (1)); the bottom plot shows the autocorrelation of the amplitude process (Eq. (3)) in blue, and the autocorrelation of the increments process in red. The autocorrelation of  $A$  (in blue in the bottom plot) reveals a mild positive memory extending over several hundred frames. The mean of the amplitudes process represents the main long-term characterization of each subcarrier in the CSI, but Eqs. (2) and (3) exclude this mean, thus Fig. 2 suggests that the amplitudes process has a residual memory beyond the subcarrier mean, producing a positive autocorrelation that persists over large lags.

The autocorrelation of the increments, in red in the bottom plot of Fig. 2, exhibits a strong one-step negative memory (approximately  $-0.4 \leq \rho_\delta(1, n) \leq -0.2, \forall n$ ). This behavior may be surprising; however, it is justified by the peculiar trend of the amplitude autocorrelation, which has an initial sharp drop followed by a long-term, non-exponential, positive autocorrelation, as shown in blue in the same plot. The computations supporting this statement are carried out in full in Sect. VI-B and Appendix A.

#### IV. DATASET CREATION

Before delving into the theoretical foundations of the previous section, we introduce the reader to the experimental contexts we worked in to assess the alignment of theory and empirical results. We bring this section forward to ensure that

Table I  
SUMMARY OF THE PROPERTIES OF THE CONSIDERED DATASETS.

|                          | ANS                          | SHARP                                      |
|--------------------------|------------------------------|--------------------------------------------|
| <b>Room</b>              | Office                       | Bedroom, laboratory, semi-anechoic chamber |
| <b>Wi-Fi standard</b>    | 802.11ax                     | 802.11ac                                   |
| <b>Tx</b>                | Asus RT-AX82U                | Netgear X4S AC2600, Asus RT-AC86U          |
| <b>Rx</b>                | Asus RT-AX82U, Asus RT-AX86U | Netgear X4S AC2600, Asus RT-AC86U          |
| <b>Monitoring device</b> | —                            | Asus RT-AC86U                              |
| <b>Inter-frame (ms)</b>  | 5, 10, 33                    | 6                                          |
| <b>BW (MHz)</b>          | 20, 40, 80                   | 80                                         |

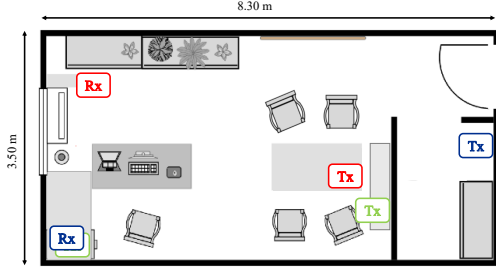


Figure 3. Office layout in the experimental testbed.

the theoretical discussion can later be carried out in parallel with its evaluation on real-world measurements.

The discussion is supported by the experiments reported in Tab. II and validated on the dataset shared by Meneghello *et al.* in [12]. The choice to use two independent datasets aims to show the robustness of the proposed approach across Wi-Fi standard versions, environments, hardware, time, and people involved in the experiments. Both datasets contain CSI's collected throughout several months, with and without Line of Sight (LoS) between the transmitter and the extractor. Since the samples were collected without artificially engineering the experimental setup, the datasets reflect real life situations and the extracted information truthfully represents the effects that human movements have on the CSI.

Table I summarizes the differences between the datasets.

#### A. ANS Dataset

All the experiments have been carried out by the Advanced Networking Systems (ANS) research group<sup>2</sup> of the University of Brescia in the office depicted in Fig. 3. All the experiments are based on long-term measures with frames separated by several milliseconds (from 5 to 33); we avoided collecting short bursts of back-to-back frames to grant the extractor, Nexmon, enough time to retrieve the CSI from each enabled receiving antenna [42]. The Tx and Rx devices are equipped with the Wi-Fi Broadcom chipset BCM43684. The Tx is always an RT-AX82U Access Points (APs) from Asus. The Rx is also an RT-AX82U for experiments with 22 ms and 33 ms inter-frame times and an RT-AX86U AP in all other cases.

<sup>2</sup><https://ans.unibs.it>

Table II  
SUMMARY OF THE EXPERIMENTS IN THE ANS DATASET. EACH SCENARIO IS IDENTIFIED BY THE 'CODE' XB.n AND THE Tx/Rx POSITION.

| Scenario      | # Frames per Exp. | Inter-frame (ms) | Tx-Rx          |
|---------------|-------------------|------------------|----------------|
| E20.0/0...7   | ~ 19000           | ~33              | R              |
| E40.0/0,1,2   | ~ 18000           | ~33              | R              |
| E80.0/0       | ~ 18000           | ~33              | R              |
| E80.0/1...9   | ~ 6000            | 10               | B(1-3), G(4-9) |
| E80.0/10,11   | ~ 12000           | 5                | B              |
| S20.1/0,1,2   | ~ 19000           | ~33              | R              |
| S40.1/0       | ~ 18000           | ~33              | R              |
| S40.2/0,1     | ~ 18000           | ~33              | R              |
| S40.3/0       | ~ 18000           | ~33              | R              |
| S40.4/0,1     | ~ 18000           | ~33              | R              |
| S40.5/0,1     | ~ 18000           | ~33              | R              |
| S80.1/0       | ~ 18000           | ~33              | R              |
| S80.1/1...8   | ~ 6000            | 10               | B(1-3), G(4-8) |
| S80.1/9,10,11 | ~ 12000           | 5                | B              |
| S80.2/0,1     | ~ 18000           | ~33              | R              |
| S80.2/2,3,4   | ~ 12000           | 5                | B              |
| S80.2/5,6     | ~ 6000            | 10               | B              |
| S80.3/0       | ~ 18000           | ~33              | R              |
| S80.4/0       | ~ 18000           | ~33              | R              |
| S80.5/0       | ~ 2700            | ~22              | R              |
| M20.4/0...3   | ~ 19000           | ~33              | R              |
| M80.1/0...7   | ~ 6000            | 10               | B(0-2), G(3-7) |
| M80.1/8,9     | ~ 12000           | 5                | B              |
| BW            | # Experiments     | # CSI samples    | Tot. Duration  |
| 20 MHz        | 15                | 248483           | ~ 150 min.     |
| 40 MHz        | 11                | 200313           | ~ 110 min.     |
| 80 MHz        | 44                | 384480           | ~ 107 min.     |

All experiments were conducted to study the behavior of CSI in real-world conditions, therefore we avoided crafting scenarios that represent edge-cases in real-life (e.g., standing perfectly still in a position). To this end, different Tx-Rx positions were used to study the impact of device placement on the behavior of CSI amplitudes. The absolute and relative positions of the Tx and Rx are classified into three configurations, represented in different colors in Fig. 3, with each set of experiments in Tab. II associated with its corresponding configuration. In the red configuration, the transmitter is placed on a desk and the receiver is in the diagonally opposite corner of the room. In the green configuration, the Tx is on a shelf about 2 m high, and the Rx is on a desk along the opposite wall. The blue configuration reuses the Rx position of the green one, but with the Tx placed in a closet room behind a drywall wall. These three configurations enable the analysis of CSI amplitude behavior with and without LoS between transmitter and receiver.

All experiments are carried out with Wi-Fi-6 (802.11ax) in the 5 GHz band with variable bandwidth (20, 40 or 80 MHz). Each scenario is identified by the 'code' XB.n where X can be E (Empty office), S (people Sitting), or M (people Moving); B is the bandwidth; n is the number of people in the room. Different experiments within the same scenario are distinguished by appending /0,1,2 to the scenario code. The last column of Tab. II specifies the positions of the Tx and Rx as red (R), green (G) or blue (B) for every set of experiments. In the experiments performed with the blue and green dispositions of the Tx/Rx, three receiving antennas are enabled, thus the number of available CSI samples triplicates in these cases.

The total amount of data is huge, as reported at the bottom of Tab. II, therefore only a few selected results are discussed.

Table III

SUMMARY OF THE EXPERIMENTS IN THE SHARP DATASET. ALL SETS INVOLVE AT LEAST ONE EXPERIMENT PER EACH ACTIVITY (EMPTY, SITTING, WALKING).

| Set | Campaign  | Environment   | LoS | Tx/Rx   |
|-----|-----------|---------------|-----|---------|
| AR1 | a-b-c-d-e | Bedroom       | Yes | Netgear |
| AR2 | a         | Bedroom       | Yes | Netgear |
| AR3 | a-b       | Bedroom       | No  | Netgear |
| AR4 | a         | Bedroom       | No  | Netgear |
| AR7 | a         | Laboratory    | Yes | Netgear |
| AR9 | a-b-c     | Semi-anechoic | Yes | Asus    |

However, all results have been verified and are supported by the analysis of the entire dataset, which is publicly available<sup>3</sup>.

### B. SHARP Dataset

The second considered dataset was used to train and test SHARP, a NN-based activity recognition tool [12]. The full dataset encompasses CSI samples collected for three different purposes: activity recognition, people counting, and person recognition. To better align with the goals of this work, we only consider data collected specifically for activity recognition. Although a detailed description of the dataset is available in [43], this section and Tab. I summarize the properties that make it a suitable complement to the ANS dataset.

The experiments involved different device setups in multi-environments: two Netgear X4S AC2600 IEEE 802.11ac routers were used in the bedroom and laboratory, whereas two Asus RT-AC86U IEEE 802.11ac routers were deployed in the semi-anechoic chamber. While for the ANS dataset the CSI is extracted at the receiver, the setup of [43] involves an Asus RT-AC86U as the monitoring device with four enabled antennas. Tx and Rx exchange traffic with 6 ms inter-frame.

Experiments performed in the laboratory and semi-anechoic chamber were made with LoS conditions, whereas both LoS and Non-Line of Sight (NLoS) conditions were set up in the bedroom, with a bookcase obstructing the propagation path.

Although the portion of the dataset dedicated to activity recognition encompasses up to seven activities, to ensure a fair comparison of the results with those obtained on the ANS dataset, we focus on the CSI samples collected in the empty environments or with a person sitting or walking. The experiments naming is consistent with the one used in [43], with the addition of a letter indicating the type of activity performed during the experiment. The naming template is therefore ARNc-a with AR standing for ‘activity recognition’; N is the set index and c indicates the campaign of acquisition; a refers to the corresponding activity and it can be E (Empty), L (Sitting), or W (Walking). For instance, the ‘code’ AR4c-L indicates the experiment belonging to the fourth set, campaign ‘c’, performed with a person sitting still. Table III summarizes the content of the dataset.

### V. CSI: AN INFORMATION ALPHABET

The processing presented in Sect. III provides insight into the properties of a single realization, i.e., one experiment, but does not shed light on what stochastic properties make different scenarios distinguishable. To address this gap, it is useful to shift from analyzing individual realizations to considering alternative representations of the CSI that can encode scenario-dependent information.

To this intent, CSI quantization is possibly the simplest way to embed the information contained in the collected samples using an arbitrarily small number of bits. This practice is already well known in the literature, although it is mostly employed to reduce the amount of information fed back to the transmitter in channel estimation applications [44]. Fewer works [45] instead delve into its applicability in contexts related to sensing. We indeed follow this approach because a quantized CSI lends itself to an interpretation as a symbol from an alphabet that carries information about the scenario in which it was collected, a representation that can be exploited for further analysis.

A first, elementary consideration is that if the CSI amplitude on each subcarrier is represented with  $q_A$  bits, the resulting alphabet has size  $2^{q_A \times N_{sc}}$ . This yields a remarkably large alphabet (or learning space): for instance, with  $q_A = 8$  and  $N_{sc} = 1024$  (as in 802.11ax with 80 MHz bandwidth), the alphabet size becomes  $2^{8192} \simeq 10^{2450}$ . Such an immense space is intriguing, as it promises very fine-grained sensing capabilities, but it is also taxingly vast for direct numerical analysis. In practice, an intuitive probabilistic approach is numerically infeasible, as the associated probabilities are vanishing. For this reason, we propose a simpler method based on the distance between CSI samples quantized over  $q_A$  bits, leaving the probabilistic analysis for future work.

We estimate the number of bits  $q_A$  required to represent the CSI amplitude as a function of the number of bits  $q_\delta$  needed to represent the increments process (Eq. (1)) with sufficient granularity. The increments are computed on the normalized CSI amplitude and are therefore bounded to the interval  $[-1, 1]$ ; however, as shown in the top plot of Fig. 2, they are rather compact around zero, consistently showing absolute values much smaller than 1. To estimate  $q_\delta$ , let  $\delta^*$  be a value such that the tails of the increments distribution outside the interval  $[-\delta^*, \delta^*]$  are negligible; then,  $q_\delta$  is the minimum value that ensures that the difference between the quantized and the empirical distributions is below a given threshold.

The increments resolution determines the representation of the amplitude changes, thus the number of bits necessary to quantize the CSI amplitude is derived as:

$$q_A = \left\lceil \log_2 \left( \frac{1}{\delta^*} \times (2^{q_\delta} + 1) \right) \right\rceil. \quad (4)$$

As discussed in Sect. VI-A, we model the increments using a T Location Scale distribution. Accordingly, we set  $\delta^* = \sigma \cdot t_\nu^{-1}(0.9985)$ , where  $t_\nu^{-1}(0.9985)$  is the 99.85th percentile of the standard Student’s T distribution with  $\nu$  degrees of freedom. With this choice, the symmetric interval  $[\mu - \delta^*, \mu + \delta^*]$  covers 99.7% of the T Location Scale distribution with parameters  $(\mu, \tau^2, \nu)$ , whose variance is  $\sigma^2 = \tau^2 \frac{\nu}{\nu-2}$ . Considering

<sup>3</sup>Zenodo: <https://doi.org/10.5281/zenodo.17225694>.

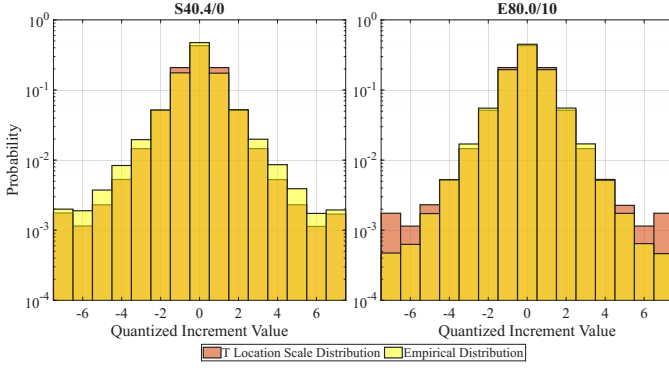


Figure 4. Comparison of the distributions of the increments (yellow) of experiments S40.4/0 (left) and E80.0/10 (right) with a T Location Scale distribution (orange). Both distributions are quantized using  $q_\delta = 4$  bits. Values falling outside the interval  $[-\delta^*, \delta^*]$  are accumulated on the lowest and highest representable values.

that the mean of the increments is zero,  $[-\delta^*, \delta^*]$  covers 99.7% of the distribution.

Extensive tests over a wide variety of experiments conducted in different scenarios suggest that  $q_\delta = 4$  and  $q_A = 9$  are appropriate values to proceed with the analysis. The interested reader can refer to [41] for a description of the methodology leading to the choice of  $q_\delta$ . Once  $q_\delta$  is fixed,  $q_A$  becomes a function of  $\delta^*$  according to Eq. (4).

Figure 4 reports the quantized distribution of the increments amplitude in a semi-logarithmic scale, measured in two very different scenarios: S40.4/0 and E80.0/10. The distribution is measured over all the subcarriers of the CSI's of the experiment and compared with a T Location Scale distribution with parameters  $(\mu, \tau^2, \nu)$  estimated by fitting the empirical distribution. Values of the increments falling outside the  $[-\delta^*, \delta^*]$  interval before quantization are accumulated on the highest or lowest representable value, rather than being discarded, to better capture the probability of large absolute values. The increments distribution is almost perfectly symmetrical, supporting the hypothesis of a locally stationary process, even with experiments lasting several minutes. Moreover, we can observe that bandwidth and frame rate do not seem to have any significant influence on the increments marginal distribution. The analysis of the remaining experiments confirms that 4 bits are sufficient to represent the increments and that the empirical distribution consistently shows heavy tails, comparable to those of a scaled Student's T distribution.

The finding that the increments distribution is invariant to bandwidth and experimental conditions is what makes the proposed quantization scheme portable across scenarios.

## VI. INCREMENTS CHARACTERIZATION

### A. Increments Distribution Analysis

Traditionally, each OFDM subcarrier is modelled as a flat-fading narrowband channel [46], with Rayleigh (NLoS) or Rician (LoS) distributions commonly used as ideal statistical models to describe the *amplitude* distribution of the complex channel gain. Accordingly, the difference between two Rayleigh or Rician random variables results in a symmetrical

distribution around zero. In the following discussion, all calculations are carried out based on the equations of the Rayleigh fading model [47], but the results are also valid for the Rician model [48].

From the sole information " $A(\cdot, n) \sim \text{Rayleigh}(\sigma)$ " nothing follows about the distribution of  $\delta(n)$ , since this is also affected by the temporal dependence between consecutive amplitude samples  $A(k-1, n)$  and  $A(k, n)$ . The following computations are valid under the standard assumption that:

$$A(k, n) \stackrel{\text{i.i.d.}}{\sim} \text{Rayleigh}(\sigma), \quad k = 1, \dots, M, \quad (5)$$

or at least that  $A(k-1, n)$  and  $A(k, n)$  are independent and with the same Rayleigh law.

Let  $X_n \sim \text{Rayleigh}(\sigma)$ , with density function

$$f_{X_n}(x) = \frac{x}{\sigma^2} e^{-\frac{x^2}{2\sigma^2}}, \quad x \geq 0$$

and Cumulative Density Function (CDF)

$$P(X_n > x) = e^{-\frac{x^2}{2\sigma^2}}. \quad (6)$$

We define  $\delta(n) = X_n - Y_n$ , with  $X_n, Y_n$  i.i.d. Rayleigh. Since  $X_n$  and  $Y_n$  follow the same law, the distribution  $\Delta(n)$  of their difference  $\delta(n)$  is symmetrical around zero.

For  $a > 0$ :

$$\begin{aligned} P(\delta(n) > a) &= P(X_n - Y_n > a) \\ &= \int_0^\infty P(X_n > a + y) \cdot f_{Y_n}(y) dy \\ &= \int_0^\infty e^{-\frac{(a+y)^2}{2\sigma^2}} \cdot \frac{y}{\sigma^2} \cdot e^{-\frac{y^2}{2\sigma^2}} dy \\ &= e^{-\frac{a^2}{2\sigma^2}} \int_0^\infty \frac{y}{\sigma^2} \cdot e^{-\frac{y^2 + ay}{\sigma^2}} dy. \end{aligned} \quad (7)$$

For  $a \rightarrow \infty$ , the dominating contribution comes from small  $y$  values. By changing the variable  $y = \frac{z}{a}$ , we obtain:

$$\begin{aligned} P(\delta(n) > a) &= e^{-\frac{a^2}{2\sigma^2}} \cdot \frac{1}{a^2 \sigma^2} \int_0^\infty z \cdot e^{-\frac{z^2}{a^2 \sigma^2}} \cdot e^{-\frac{z}{\sigma^2}} dz \\ &\sim \frac{\sigma^2}{a^2} \cdot e^{-\frac{a^2}{2\sigma^2}}. \end{aligned}$$

For symmetry:

$$P(\delta(n) < -a) = P(\delta(n) > a) \sim \frac{\sigma^2}{a^2} \cdot e^{-\frac{a^2}{2\sigma^2}}. \quad (8)$$

According to Eq. (5), all subcarriers in any collection obtained under the same LoS conditions have the same amplitude distribution and, consequently, the same increments distribution  $\Delta(n)$ . Thus, to collect more samples and better characterize the model, we merge all one-step increments computed between any pair of consecutive CSI's on any subcarrier within the same experiment. Through this operation, we obtain a single increments distribution  $\Delta$  per experiment.

In [19] we assumed it to be described by a simple Gaussian model, a simplifying choice justified by the symmetry around zero of the increments distribution. In fact, Eq. (8) highlights that the distribution of the increments has heavier tails than a Gaussian  $\mathcal{N}$  with the same variance.

Since the Gaussian distribution fails to capture the observed tail behavior, an accurate model requires a more general

symmetric distribution capable of modelling excess kurtosis. To address this, we consider two additional candidates: the Logistic and Student's T distributions. Both preserve the symmetry property of the Gaussian distribution, while allowing a higher kurtosis that induces heavier tails. Specifically, the Logistic distribution preserves the exponential tail behavior expected from theoretical evaluations on the increments, while the Student's T distribution has significantly heavier, polynomial tails. The Logistic distribution is parametrized by location and scale, like the Gaussian, whereas Student's T distribution is characterized solely by its degrees of freedom  $\nu$ .

The selection of the Student's T distribution can also be motivated by the underlying variability of the amplitude evolution process. In the derivation of Eq. (8), we assume that the scale parameter  $\sigma$  is fixed over time and across subcarriers, due to the i.i.d. hypothesis. In realistic environments, instead, this assumption is not satisfied: human motion, AGC effects, and changes in the environment geometry, for instance, introduce fluctuations in the Rayleigh-distributed CSI amplitude. A more accurate amplitude model is therefore

$$A(k, n) | S_k \sim \text{Rayleigh}(\sigma_k),$$

where  $S_k$  is the propagation state that affects the amplitude of the  $k$ -th CSI and  $\sigma_k$  varies in time.

Consequently, in the long term, the increments are better modelled by a *mixture* of Rayleigh-difference distributions with varying  $\sigma$ , rather than by a fixed-scale Rayleigh-difference. Such mixtures are generally heavier-tailed than single-scale models: rare, high-variance states (i.e. more dynamic environmental conditions) increase the probability of larger increments, while low-variance ones keep the increments concentrated near zero.

The Student's T family is chosen as an adaptable model for this scale mixture mechanism and is consistent with the observed heavy tails across all experiments. This shows that the distribution is a mathematically consistent approximation for the CSI amplitude increments and is not merely an empirical fit. Specifically, we employ the three-parameter generalization of Student's T distribution: the T Location Scale distribution  $ls_T(\mu, \tau^2, \nu)$ . This reduces to the standard Student's T for  $ls_T(\mu = 0, \tau^2 = 1, \nu)$ , and converges to the Normal  $\mathcal{N}(\mu, \tau^2)$  for  $\nu \rightarrow \infty$ . To ensure finite mean and variance when fitting the increments distribution, we enforce  $\nu > 2$ .

Fig. 5 shows two examples of fitting of the three candidate distributions and compares them to the distribution obtained from the difference of two i.i.d. Rayleigh variables. Such distribution has the same variance as the Gaussian. Indeed, we observe that its tails are heavier than those of the Gaussian distribution, but they do not accurately represent the empirical increments distribution. Regardless of the scenario (five people sitting versus the empty room) and the channel bandwidth, an exponentially-tailed distribution is clearly too coarse an approximation for the increments, as it fails to capture either the mode or the tails with sufficient accuracy. In contrast, the T Location Scale distribution captures both features in detail, providing the most reliable fit.

To evaluate the goodness of fit of the three candidate distributions, for each experiment, we fit the distributions to

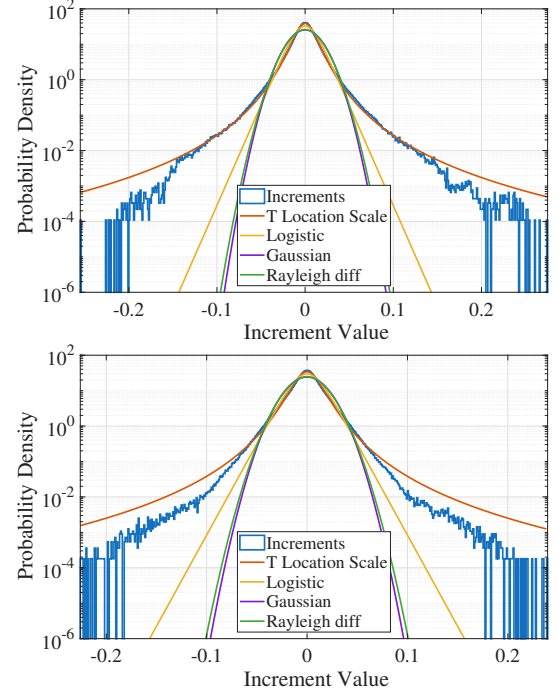


Figure 5. Fitting of the distributions of the increments of experiments S40.5/1 and E80.0/3 with the three candidate distributions.

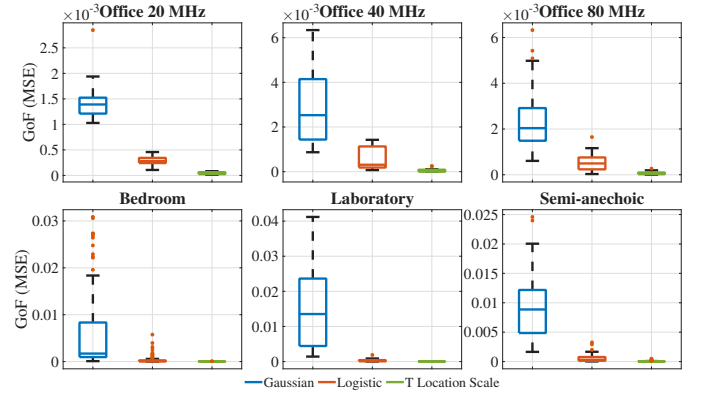


Figure 6. Aggregated representation of the MSE between the CDF of the three candidate distributions and the CDF of the increments of every experiment in the dataset. Experiments are grouped by the pair <room, bandwidth>. The error values obtained from the Gaussian distribution fit to the SHARP dataset (bottom row) are one order of magnitude larger than those for the Logistic and T Location Scale distributions, shrinking their scale on the y-axis.

the observed increments, estimate the corresponding parameters, and generate the CDF of the fitted distribution. We then compute the Mean Squared Error (MSE) between this CDF and the empirical CDF of the observed increments. The resulting MSE values obtained across all experiments are grouped by room and bandwidth and represented in the boxplots in Fig. 6, which shows that the T Location Scale distribution consistently yields by far the lowest MSE with a very small scattering.

The Office-80 MHz and Bedroom cases require further clarification: aggregating the MSE values combines results obtained from data collected under heterogeneous experimental conditions. Compared to the 20 and 40 MHz setups in the

Table IV  
MAXIMUM STANDARD DEVIATION FOR EACH CANDIDATE DISTRIBUTION  
IDENTIFIED AFTER FITTING ON ALL EXPERIMENTS.

| $\sigma$ (max)   | ANS<br>20 MHz | ANS<br>40 MHz | ANS<br>80 MHz | SHARP<br>80 MHz |
|------------------|---------------|---------------|---------------|-----------------|
| Increments       | 0.0164        | 0.0443        | 0.0520        | 0.0294          |
| Gaussian         | 0.0164        | 0.0443        | 0.0520        | 0.0294          |
| Logistic         | 0.0157        | 0.0390        | 0.0478        | 0.0279          |
| T Location Scale | 0.0171        | 0.0447        | 0.0575        | 0.0308          |

ANS dataset, the 80 MHz-bandwidth experiments are more diverse, encompassing also additional experiments with much lower inter-frame time, as well as alternative Rx-Tx positions. Similarly, the Bedroom setup in the SHARP dataset merges data collected with and without LoS. Nevertheless, the T Location Scale distribution consistently fits with precision the increments distribution, indicating that device placement and inter-frame time between consecutive CSI samples do not affect their characterization.

To confirm that the T Location Scale is the best-fitting distribution, we also compute the log-likelihood of the fitted Probability Distribution Functions (PDFs). For a probability density function  $f(\delta(i); \theta)$  with parameters  $\theta$ , the log-likelihood is defined as

$$\text{Log}L = \sum_{i=1}^n \log(f(\delta(i); \theta)).$$

This allows the computation of the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) for  $k$  parameters and  $m$  increments:

$$\text{AIC} = 2 \cdot k - 2 \cdot \text{Log}L \quad (9)$$

$$\text{BIC} = k \cdot \log(m) - 2 \cdot \text{Log}L \quad (10)$$

Lower values of AIC and BIC indicate a better trade-off between model fit and complexity, as these criteria penalize models with more parameters while rewarding higher likelihood. In particular, the BIC penalizes model complexity more strongly as the dataset size  $m$  increases, resulting in a stricter criterion compared to the AIC. AIC and BIC are not absolute measures of model quality; they are relative metrics that allow *comparison* of candidate models fitted to the same data. The T Location Scale distribution consistently emerges as the best-fitting model for the empirical increments distribution across all bandwidths and experiments.

Table IV presents the maximum standard deviation  $\sigma$  of the increments – computed using the unbiased estimator – and the three candidate distributions fitted to them. As expected, the Gaussian distribution replicates the empirical standard deviation, because the estimated value of the  $\sigma$  parameter is the square root of the unbiased estimate of the variance. On the contrary, the Logistic and T Location Scale distributions yield slightly different values, respectively slightly under and overestimating the standard deviation of the increments.

For the T Location Scale, especially, we highlight how it does not rely solely on the scale parameter to capture the variability of the data but it also accounts for tail behavior. In fact, the T Location Scale characterizes the increments

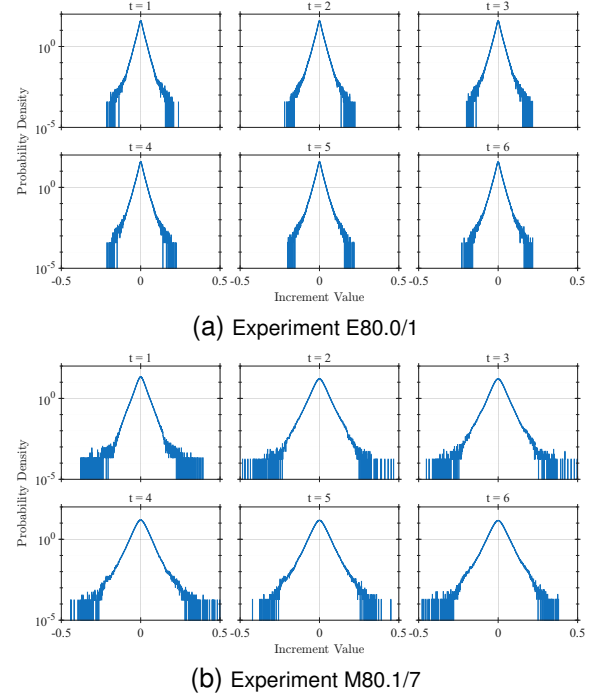


Figure 7.  $t$ -step increments distributions for two experiments,  $t = 1 \dots 6$ .

variability *mainly* through its tail parameter  $\nu$ . The implication of heavy tails distribution on the increments is relevant, as they are associated to either long-term dependence or non-strictly stationary behavior of the observed process; wide-sense, local stationary behavior may be preserved, as we already observed, but strict or long-term stationary behavior may be absent, definitely influencing learning processes and sensing strategies.

The discussion so far was only focused on one-step increments, but it can be extended to  $t$ -step increments:

$$\delta^t(k, n) = A(k, n) - A(k - t, n) \quad (11)$$

where  $k \in [t, M]$  and  $n$  is the usual OFDM subcarrier index. The considerations made about one-step increments hold for  $\delta^t$  as well. Figure 7 exemplifies how the distributions of the increments for  $t = 1 \dots 6$  remain substantially unaltered regardless of  $t$ . Moreover, Fig. 7 helps to visually grasp the increasing variability of the increments values in more dynamic environments, as the tail differences between scenarios E and M stand out.

The statistical properties of the  $t$ -step increments distribution remain the same even for  $t > 1$ , as shown in Fig. 8. Regardless of bandwidth, considered Rx antenna, or environment, a T Location Scale distribution with a stable range of degrees of freedom well describes the increments. Each enabled Rx antenna contributes one  $\nu$  value to the statistics reported in Fig. 8. For every collection of CSI's associated with each antenna, the standard deviation of  $\nu$  across time steps  $t = 1, \dots, 6$  never exceeds 0.8944 in the ANS dataset and 3.202 in the SHARP dataset. Overall, the standard deviation is zero for 49.11% of CSI collections, while only 6.85% exhibit a standard deviation greater than 1.

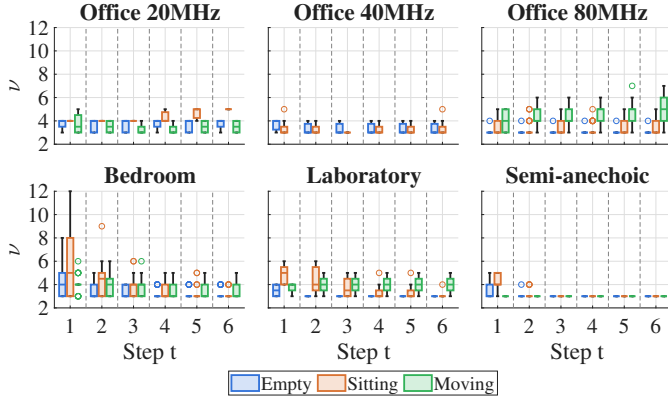


Figure 8. Variability of  $\nu$  values identified while fitting the T Location Scale distribution to the  $t$ -step increments of all experiments, with  $t = 1 \dots 6$ . Experiments are grouped by the pair <room, bandwidth>. Each enabled Rx antenna contributes with one  $\nu$  value.

The stability of  $\nu$  values across  $t$  indicates that the tail behavior of the increments distribution is preserved at different granularities. This suggests that the identified T Location Scale model provides a robust characterization of the increments regardless of the chosen time step. To the best of our knowledge this is the first time that heavy tails are documented in the subcarriers amplitude of OFDM-based CSI estimates; while this has no effect on transmission or equalization, it may have important consequences in sensing.

### B. Autocorrelation Analysis

The autocorrelation analysis helps describe the memory structure of the CSI process and validates the departure from i.i.d. assumptions that are considered standard in the literature. Moreover, it provides an analytical explanation for the counter-intuitive behavior of the increments autocorrelation observed empirically (see Fig. 2). In fact, the example autocorrelation of the amplitudes ( $\rho_A$ ) in the bottom plot of Fig. 2 reveals residual memory in the amplitude process beyond the mean, as introduced in Sect. III. This residual memory also affects the autocorrelation of the increments ( $\rho_\delta$ ), producing the observed, counter-intuitive one-step negative correlation.

Applying the definition of auto-covariance provided in Eq. (2) to the increments, we have:

$$\begin{aligned} R_\delta(\tau, n) &= \mathbb{E}[\delta(k, n) \cdot \delta(k - \tau, n)] - \mu_{\delta(\cdot, n)}^2 \\ &= \mathbb{E}[(A(k, n) - A(k - 1, n)) \\ &\quad \cdot (A(k - \tau, n) - A(k - 1 - \tau, n))], \end{aligned}$$

knowing that  $\mu_{\delta(\cdot, n)} = \mathbb{E}[\delta(n)] = 0$ . Developing the product of  $A$  and leveraging the linearity of the  $\mathbb{E}[\cdot]$  operator, we obtain:

$$\begin{aligned} R_\delta(\tau, n) &= \mathbb{E}[(A(k, n) \cdot A(k - \tau, n)) \\ &\quad - \mathbb{E}[(A(k, n) \cdot A(k - 1 - \tau, n))] \\ &\quad - \mathbb{E}[(A(k - 1, n) \cdot A(k - \tau, n))] \\ &\quad + \mathbb{E}[(A(k - 1, n) \cdot A(k - 1 - \tau, n))] \\ &= 2R_A(\tau, n) - R_A(\tau + 1, n) - R_A(\tau - 1, n) \end{aligned} \quad (12)$$

For  $\tau = 0, 1$ , we have:

$$R_\delta(0, n) = 2R_A(0, n) - 2R_A(1, n) \quad (13)$$

$$R_\delta(1, n) = 2R_A(1, n) - R_A(2, n) - R_A(0, n), \quad (14)$$

which, plugged in Eq. (3), give an autocorrelation  $\rho_\delta$  of:

$$\rho_\delta(0, n) = 1 \quad (15)$$

$$\rho_\delta(1, n) = \frac{2\rho_A(1, n) - \rho_A(2, n) - 1}{2 - 2\rho_A(1, n)}. \quad (16)$$

The denominator of Eq. (16) is always positive, assuming  $\rho_A(1, n) < 1$ ; the numerator is instead typically negative, since the contribution of the unit autocorrelation  $\rho_A(0, n)$  is larger than higher-lag terms. This is what we observe in the empirical measures (see Fig. 2), with the theoretical analysis confirming the observation. The full derivation of Eq. (16) is reported in Appendix A.

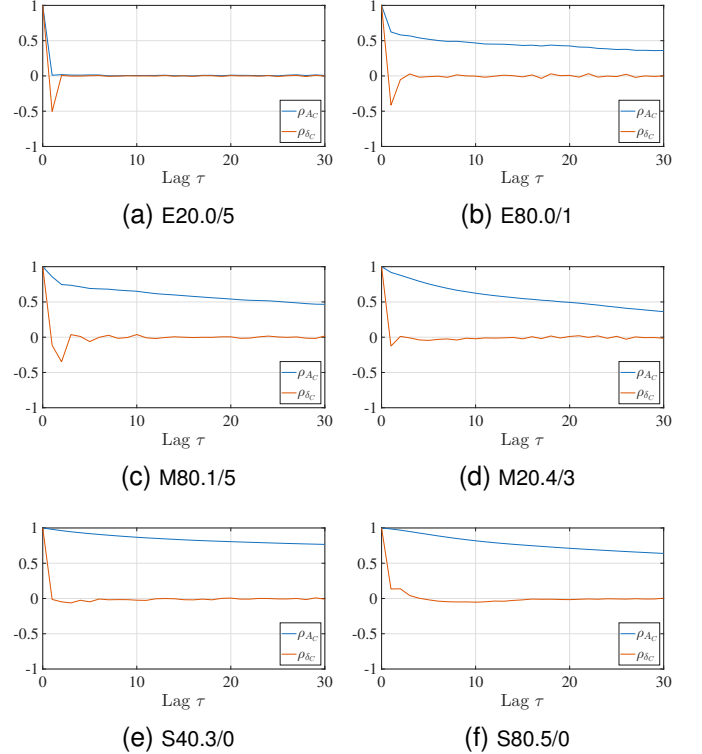


Figure 9. Zoom on lags  $\tau = 0 \dots 30$  of the average  $\rho_A$  and  $\rho_\delta$  computed across all subcarriers for six selected experiments.

Figure 9 reports additional examples of the amplitude (in blue) and increments (in red) autocorrelations computed in different experiments. The figure highlights the relationship between a sharp drop in the initial lags of  $\rho_A$  and the presence of a negative value of  $\rho_\delta(1, n)$ , and allows further insight. In Figs. 9a, 9b, and 9c, where  $\rho_A$  exhibits a visible drop, the negative value of  $\rho_\delta$  at  $\tau = 1$  is evident; on the contrary, in Figs. 9d, 9e, and 9f, the slower decay of the  $\rho_A$  is associated with a memoryless increments behavior. Additionally, the amplitude autocorrelation exhibits a slow and non-exponential decay persisting well above zero over large lags.

The one-step memory in the increments is analytically explained by Eq. (16) as a consequence of the amplitude

autocorrelation structure rather than independent evidence of higher-order memory in the increments themselves. The slow, non-exponential decay of  $\rho_A$ , instead, highlights the non-Markovian character of the amplitude process with respect to the experimentally measured increment process.

## VII. A DISTANCE METRIC FOR AMBIENT SENSING

An interesting question is whether the stochastic features of the CSI evolution process analyzed in the previous sections can inform JCAS methods. To this end, we devise a simple metric as an example of how even basic knowledge of the increments distribution properties that drive CSI quantization can already provide distinctive information about the environment where the CSI's were collected.

After the normalization and quantization process described in Sect. V and in [41], the amplitude of a CSI  $A(k, \cdot)$  becomes a vector of  $N_{sc}$  integer numbers represented over  $q_A = 9$  bits.

We define the distance between two arbitrary CSI samples  $i$  and  $j$ , independently from the experiment, as

$$W_D(A(i), A(j)) = \sum_{n=-\frac{N_{sc}}{2}}^{\frac{N_{sc}}{2}-1} |A(i, n) - A(j, n)| \quad (17)$$

assuming that the two CSI's have the same number of subcarriers. Equation (17) can be used to estimate and quantify the difference between two CSI samples and, consequently, to determine whether they belong to different scenarios, under the assumption that CSI's collected within the same scenario are more similar and therefore closer to one another.

To maintain a reasonably simple approach, we define the representative CSI of a specific experiment  $C(\cdot, \cdot)$  as the average of the CSI amplitudes collected during that experiment:

$$A^* = \frac{1}{M} \sum_{k=1}^M A(k). \quad (18)$$

This way, the distance  $W_D(A(k), A_j^*)$  indicates the “similarity” of the generic  $k$ -th CSI of experiment  $C(\cdot, \cdot)$  to an experiment  $C_j(\cdot, \cdot)$  conducted in the scenario  $S_j$ . The local wide-sense stationarity assumption is the hypothesis that ensures that  $A^*$  is representative of the experiment. If the process were not stationary within an experiment, averaging would be incoherent.

Finally, we can obtain a compact indicator of the similarity of two experiments by computing the average distance of all CSI's of one experiment from the average CSI of the other:

$$\overline{W_D(A_i, A_j^*)} = \frac{1}{M} \sum_{k=1}^M W_D(A_i(k), A_j^*) \quad (19)$$

Clearly,  $\overline{W_D(A_i, A_j^*)} \neq \overline{W_D(A_j, A_i^*)}$ , even if we expect the two values to be similar.

Tables V, VI, VII, and VIII report the metric  $\overline{W_D(A_i, A_j^*)}$  computed on  $W_D(A_i, A_j^*)$  values normalized to 1 – i.e., divided by  $N_{sc} \cdot (2^{q_A} - 1)$  – for selected experiments with 20, 40, and 80 MHz bandwidth from the ANS dataset and experiments from the laboratory setup in the SHARP dataset. Since all numbers are in  $[0, 1)$ , we only report four decimal

digits after the dot for conciseness (e.g., 0.0491  $\rightarrow$  0491). This being the average distance, we cannot expect structured and rich information; however, interesting observations can already be derived.

In all cases, the minimum average distance of the CSI samples  $A_i$  of experiment  $i$  is the one computed from the average CSI  $A_i^*$  of the experiment  $i$  itself. The only exception is experiment E80.0/4, which has the same  $\overline{W_D}$  from its own  $A_i^*$  and from that of E80.0/8: both experiments were conducted in the same empty-room conditions within minutes of each other, yielding extremely similar CSI samples. Thus, it is not surprising, but indeed encouraging, that these two cannot be distinguished easily.

Regardless of the bandwidth, the distance between experiments conducted in different scenarios stands out, as it is generally higher than that computed between experiments from the same scenario. The empty room is easy to recognize, with CSI samples closer to others collected in the same environmental conditions than to those impacted by a person's presence or movements.

Among the four tables, Tab. VII is especially noteworthy because it highlights the impact of different Tx placements on CSI distance. In the empty-room experiments, the distance between experiments E80.0/4 and E80.0/8, both collected with the green configuration, is mutually very similar; on the contrary, both have a large distance from E80.0/1 that was collected in the blue setup (see Fig. 3). The result is not surprising as it simply confirms theory, but it also highlights that great attention is needed to perform CSI-based sensing. The impact of Tx placement is less evident when a person is sitting or moving around the room: these scenarios have an intrinsically higher CSI variability, making it hard to attribute a larger  $\overline{W_D}$  to a different Tx placement rather than to human activity.

These considerations are dataset-independent: in Tab. VIII every experiment from the SHARP dataset remains consistently closer to its representative CSI than to other experiments. This result confirms that the behavior of the proposed metric does not depend on a specific experimental setup or on the devices used to collect the data; on the contrary, it highlights its great adaptability to other scenarios, channel bandwidths, and even 802.11 standard versions.

To better grasp the effect of environment variations on  $W_D$ , Fig. 10 shows the distribution of all normalized  $W_D(A_i, A_i^*)$  of a selection of experiments from their own average CSI. The four subplots confirm the statements on the average  $W_D$ : CSI samples collected in an empty room are very similar to each other, yielding consistently small distances from their average CSI regardless of the variability of the two datasets. On the opposite end, the increased dynamic due to human presence and activity results in larger distances and complex spreads of the distance distribution. This observation implies that similitude is not a good means of classification to discriminate non-static situations; a simple consideration that, however, does have consequences on any sensing and recognition algorithm, including those based on AI techniques. Furthermore, as documented in [19], the analysis of the distance from the expected value represents a powerful and efficient means

Table V  
 $\overline{W_D(A_i, A_j^*)}$  FOR SELECTED ANS EXPERIMENTS WITH 20 MHz-BW; IN BOLD, THE MINIMUM DISTANCE FOR EACH EXPERIMENT (COLUMN).

|       | A       |             |             |             |             |             |             |             |             |             |
|-------|---------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|       | E20.0/0 | E20.0/1     | E20.0/6     | S20.1/0     | S20.1/1     | S20.1/2     | M20.4/0     | M20.4/1     | M20.4/2     | M20.4/3     |
| $A^*$ | E20.0/0 | <b>0085</b> | 0199        | 0709        | 0758        | 0855        | 1255        | 1924        | 1554        | 0935        |
|       | E20.0/1 | 0203        | <b>0069</b> | 0838        | 0886        | 0926        | 1035        | 1410        | 2032        | 1078        |
|       | E20.0/6 | 0709        | 0837        | <b>0087</b> | 0851        | 0821        | 0899        | 1516        | 2230        | 1267        |
|       | S20.1/0 | 0686        | 0872        | 0872        | <b>0185</b> | 0327        | 0294        | 0868        | 1449        | 1278        |
|       | S20.1/1 | 0722        | 0900        | 0769        | 0285        | <b>0240</b> | 0266        | 0948        | 1569        | 1339        |
|       | S20.1/2 | 0838        | 1021        | 0865        | 0295        | 0292        | <b>0202</b> | 0860        | 1422        | 1254        |
|       | M20.4/0 | 1201        | 1357        | 1449        | 0793        | 0895        | 0793        | <b>0450</b> | 1133        | 0890        |
|       | M20.4/1 | 1848        | 1960        | 2161        | 1371        | 1498        | 1336        | 1079        | <b>0556</b> | 0748        |
|       | M20.4/2 | 1487        | 1594        | 1867        | 1193        | 1266        | 1173        | 0860        | 0770        | <b>0533</b> |
|       | M20.4/3 | 0894        | 1037        | 1227        | 0596        | 0660        | 0591        | 0602        | 1168        | <b>0322</b> |

Table VI  
 $\overline{W_D(A_i, A_j^*)}$  FOR SELECTED ANS EXPERIMENTS WITH 40 MHz-BW; IN BOLD, THE MINIMUM DISTANCE FOR EACH EXPERIMENT (COLUMN).

|       | A       |             |             |             |             |             |             |             |             |             |
|-------|---------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|       | E40.0/0 | E40.0/1     | E40.0/2     | S40.1/0     | S40.2/0     | S40.3/0     | S40.4/0     | S40.4/1     | S40.5/0     | S40.5/1     |
| $A^*$ | E40.0/0 | <b>0050</b> | 0133        | 0160        | 2772        | 0931        | 1499        | 1378        | 1441        | 1901        |
|       | E40.0/1 | 0096        | <b>0098</b> | 0226        | 2715        | 0872        | 1475        | 1334        | 1391        | 1853        |
|       | E40.0/2 | 0161        | 0251        | <b>0050</b> | 2901        | 1043        | 1576        | 1496        | 1542        | 2000        |
|       | S40.1/0 | 2747        | 2693        | 2872        | <b>0361</b> | 2266        | 2747        | 2519        | 2485        | 2489        |
|       | S40.2/0 | 0858        | 0796        | 0990        | 2259        | <b>0416</b> | 1262        | 1218        | 1294        | 1538        |
|       | S40.3/0 | 1437        | 1404        | 1528        | 2735        | 1241        | <b>0497</b> | 1461        | 1891        | 1823        |
|       | S40.4/0 | 1147        | 1084        | 1293        | 2364        | 1012        | 1305        | <b>0875</b> | 1166        | 1535        |
|       | S40.4/1 | 1157        | 1085        | 1283        | 2258        | 0978        | 1722        | 1067        | <b>0983</b> | 1601        |
|       | S40.5/0 | 1560        | 1494        | 1678        | 2282        | 1079        | 1552        | 1339        | 1480        | <b>1130</b> |
|       | S40.5/1 | 1348        | 1301        | 1468        | 2565        | 1112        | 0952        | 1269        | 1634        | <b>0720</b> |

Table VII  
 $\overline{W_D(A_i, A_j^*)}$  FOR SELECTED ANS EXPERIMENTS WITH 80 MHz-BW; IN BOLD, THE MINIMUM DISTANCE FOR EACH EXPERIMENT (COLUMN).

|       | A       |             |             |             |             |             |             |             |             |             |
|-------|---------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|       | E80.0/1 | E80.0/4     | E80.0/8     | S80.1/5     | S80.1/6     | S80.1/8     | M80.1/2     | M80.1/3     | M80.1/4     | M80.1/6     |
| $A^*$ | E80.0/1 | <b>0101</b> | 2101        | 2103        | 1576        | 1679        | 2010        | 1123        | 1947        | 1528        |
|       | E80.0/4 | 2102        | <b>0088</b> | 0089        | 1352        | 1167        | 0831        | 2077        | 1099        | 1744        |
|       | E80.0/8 | 2103        | <b>0088</b> | <b>0088</b> | 1356        | 1168        | 0834        | 2078        | 1100        | 1748        |
|       | S80.1/5 | 1557        | 1330        | 1335        | <b>0295</b> | 0549        | 1020        | 1432        | 1150        | 1061        |
|       | S80.1/6 | 1661        | 1127        | 1128        | 0551        | <b>0280</b> | 1095        | 1536        | 1041        | 1395        |
|       | S80.1/8 | 1961        | 0750        | 0753        | 0987        | 1072        | <b>0336</b> | 1725        | 0975        | 1208        |
|       | M80.1/2 | 0919        | 1977        | 1978        | 1362        | 1474        | 1660        | <b>0542</b> | 1640        | 1232        |
|       | M80.1/3 | 1882        | 0930        | 0931        | 1047        | 0918        | 0881        | 1625        | <b>0545</b> | 1399        |
|       | M80.1/4 | 1473        | 1689        | 1693        | 1028        | 1375        | 1209        | 1339        | 1423        | <b>0371</b> |
|       | M80.1/6 | 1423        | 1740        | 1744        | 1059        | 1424        | 1248        | 1322        | 1482        | <b>0403</b> |

Table VIII  
 $\overline{W_D(A_i, A_j^*)}$  FOR SHARP-LABORATORY EXPERIMENTS; IN BOLD, THE MINIMUM DISTANCE FOR EACH EXPERIMENT (COLUMN).

|       | A       |             |             |             |
|-------|---------|-------------|-------------|-------------|
|       | AR7a.E  | AR7a.L1     | AR7a.L2     | AR7a.W      |
| $A^*$ | AR7a.E  | <b>0017</b> | 1725        | 1127        |
|       | AR7a.L1 | 1633        | <b>0511</b> | 1504        |
|       | AR7a.L2 | 0997        | 1486        | <b>0566</b> |
|       | AR7a.W  | 0546        | 1466        | 0828        |

to identify outliers due, for instance, to errors in the CSI extraction tool.

Until this point, only CSI samples collected on one antenna (Antenna 0) were presented and discussed in this section, but all experiments with Tx/Rx positions marked in blue and green in Fig. 3 had three receiving antennas enabled. Similarly, CSI collections from the SHARP dataset were obtained using all four antennas available on the monitoring device. Focusing on the computation of the  $W_D$  distance between CSI samples collected on the same antenna and their corresponding average  $A^*$ , the resulting distance distributions for ANS experiments E80.0/6, S80.1/8, and M80.1/4 are shown in the left column of Fig. 11. The right column, instead, shows the distributions

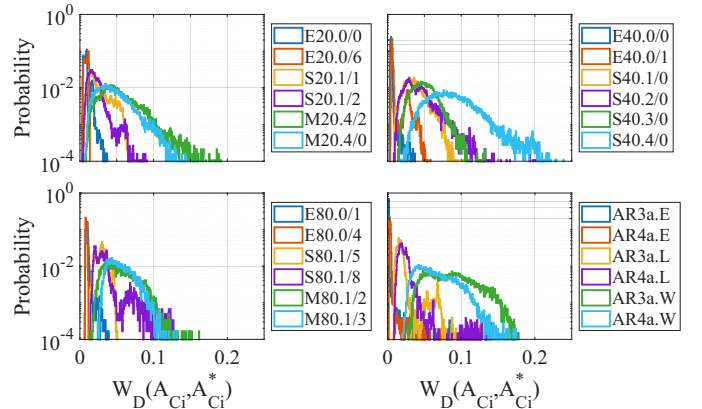


Figure 10. Distribution of normalized  $W_D(A_i, A_i^*)$  for some selected experiments from both datasets.

obtained from SHARP experiments AR9a.E, AR9a.L1, and AR9a.W1. The plots reveal a strong similarity among CSI samples collected on the same antenna in the empty scenario, confirming that a static environment enforces only minimal deviations from the average CSI structure. In contrast, human presence makes CSI's more variable, increasing the distance

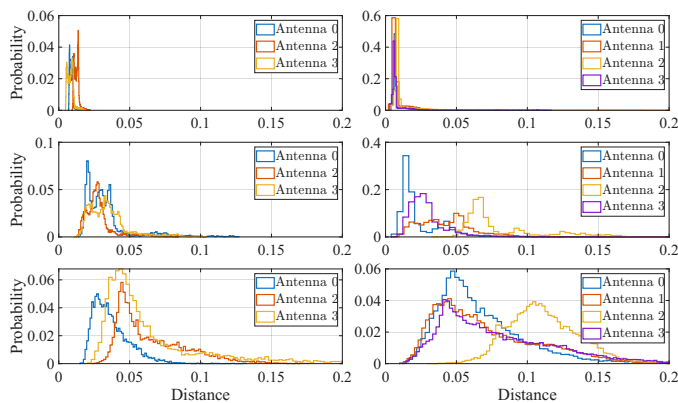


Figure 11. Distributions of  $W_D(A_i, A_i^*)$  for selected experiments. Left column: E80.0/6 (top), S80.1/8 (middle), and M80.1/4 (bottom). Right column: AR9a.E (top), AR9a.L1 (middle), AR9a.W1 (bottom).

from the average. These trends are consistent across all receiving antennas in both datasets and coherent with the discussion on Fig. 10.

## VIII. A SIMPLE ENVIRONMENT CLASSIFICATION ALGORITHM

The focus of this work is on the analysis of fundamental properties of CSI amplitudes with the goal of complementing and informing sensing and classification methods. However, it is clear that, in the context of IoT, computationally light methodologies to detect human presence are preferable to resource-consuming tools, thus we introduce a simple distance-based classification algorithm and compare its performance with a state of the art AI algorithm and with an ML one based on an SVM. In Sect. VII we have introduced the average  $W_D$  as a stochastically-informed, elementary metric that highlights environmental differences due to variable levels of dynamics. Building on the already-discussed preliminary results, we propose a simple algorithm that leverages  $W_D$  to discriminate empty from non-empty rooms: if the average distance of an experiment from the average CSI collected in the empty room is smaller than a threshold  $\mu_{th}$ , the room is classified as empty, otherwise as occupied.

There are many possible approaches to calculate  $\mu_{th}$  and also to optimize its computation, but the goal here is just to show how computationally inexpensive techniques can support CSI sensing. Thus we take one third of the CSI samples collected in the empty room as representative of the empty scenario ( $M_T$  samples) and use them to compute the threshold  $\mu_{th} = \mu(W_D) + 3 \cdot \sigma(W_D)$ . This procedure is repeated separately for all experiment configurations to ensure robustness across channel bandwidths and LoS conditions in the ANS dataset; for the SHARP dataset, instead, experiments are grouped by the pair  $\langle \text{room, LoS conditions} \rangle$ , since all were conducted using an 80 MHz bandwidth. To ensure the contributions of all antennas are accounted for, the room occupancy detection algorithm implements the classification separately on the CSIs collected from each enabled antenna; then, a majority voting approach on the classification outcomes

Table IX  
ACCURACY AND F1-SCORE FOR ENVIRONMENT CLASSIFICATION.  
EXPERIMENTS ARE GROUPED BY THE TUPLE  $\langle \text{DATASET, ROOM, BANDWIDTH, SETUP, INTERFRAME} \rangle$ .

| ID  | Dataset             | BW     | Setup | Inter frame | Accuracy | F1-Score |
|-----|---------------------|--------|-------|-------------|----------|----------|
| (a) | ANS Office          | 20 MHz | R     | 33 ms       | 100%     | 100%     |
| (b) | ANS Office          | 40 MHz | R     | 33 ms       | 100%     | 100%     |
| (c) | ANS Office          | 80 MHz | R     | 33 ms       | 100%     | 100%     |
| (d) | ANS Office          | 80 MHz | G     | 10 ms       | 100%     | 100%     |
| (e) | ANS Office          | 80 MHz | B     | 10 ms       | 75%      | 80%      |
| (f) | ANS Office          | 80 MHz | B     | 5 ms        | 90%      | 93.33%   |
| (g) | SHARP Bedroom       | 80 MHz | LoS   | 6 ms        | 100%     | 100%     |
| (h) | SHARP Bedroom       | 80 MHz | NLoS  | 6 ms        | 66.67%   | 66.67%   |
| (i) | SHARP Laboratory    | 80 MHz | LoS   | 6 ms        | 100%     | 100%     |
| (j) | SHARP Semi-anechoic | 80 MHz | LoS   | 6 ms        | 76.47%   | 83.33%   |

Table X  
ACCURACY AND F1-SCORE FOR SVM-BASED CLASSIFICATION USING  $M_T$  CSI SAMPLES FROM ‘E’ AND  $M_T$  FROM ‘S’ AND ‘M’. EXPERIMENTS ARE GROUPED BY THE TUPLE  $\langle \text{DATASET, ROOM, BANDWIDTH, SETUP, INTERFRAME} \rangle$ .

| ID  | Dataset             | BW     | Setup | Inter frame | Accuracy | F1-Score |
|-----|---------------------|--------|-------|-------------|----------|----------|
| (a) | ANS Office          | 20 MHz | R     | 33 ms       | 100%     | 100%     |
| (b) | ANS Office          | 40 MHz | R     | 33 ms       | 80.58%   | 85.65%   |
| (c) | ANS Office          | 80 MHz | R     | 33 ms       | 99.97%   | 99.98%   |
| (d) | ANS Office          | 80 MHz | G     | 10 ms       | 100%     | 100%     |
| (e) | ANS Office          | 80 MHz | B     | 10 ms       | 99.93%   | 99.96%   |
| (f) | ANS Office          | 80 MHz | B     | 5 ms        | 99.47%   | 99.70%   |
| (g) | SHARP Bedroom       | 80 MHz | LoS   | 6 ms        | 83.68%   | 88.90%   |
| (h) | SHARP Bedroom       | 80 MHz | NLoS  | 6 ms        | 71.21%   | 83.15%   |
| (i) | SHARP Laboratory    | 80 MHz | LoS   | 6 ms        | 100%     | 100%     |
| (j) | SHARP Semi-anechoic | 80 MHz | LoS   | 6 ms        | 93.76%   | 96.08%   |

determines the final labelling of the environment as empty or occupied.

The performance of the algorithm is summarized in Tab. IX, which shows the values of accuracy and F1-score obtained by the proposed algorithm on the two datasets. Most scenarios are correctly classified with over 90% accuracy, assessing the adaptability and stability of the proposed metric. In fact, only the experiments of the ANS dataset lacking LoS and those of the SHARP dataset conducted in the bedroom have lower accuracy percentages. However, this behavior is expected: the configuration these experiments reduces the impact human motion can have on the CSI. In the blue (B) setup in the ANS dataset (see Fig. 3) a wall obstructs the propagation path between the Tx and the Rx. As for the experiments from the SHARP dataset, according to Fig. 5 of [12], in the NLoS bedroom scenario a bookshelf shields the area where the person is sitting or moving from the area where the Tx and Rx are placed. These scenarios hinder the distinction of the alterations to the CSI imposed by the obstruction of the propagation path from those due to the human body. This results in an increased difficulty in detecting static human presence and multiple misclassifications of the ‘sitting’ experiments as ‘empty’. This outcome is strictly related to the real-world conditions where the data were collected: the experiments cannot be rigidly controlled and some scenarios “overlap”.

### A. Comparison with SVM-based Classification

To frame our method in the context of IoT and embedded systems, we compare its performance to that of a linear SVM.

The SVM – implemented in Matlab through the `fitcecoc` function – works as a supervised classifier taking as input the quantized CSI samples collected in the empty and occupied room; then, it identifies the hyperplanes that maximize the margin between CSI's corresponding to the two situations and learns the deriving decision function to perform classification. To prevent information leak and classification bias, as well as to preserve the natural flow of the CSI time sequence, we avoid randomly shuffling the CSI samples when generating the training and testing set. Instead, the first part of the experiments made in each situation contributes to the training set; the remainder belongs to the testing set. Overall, the training set consists of  $T_s$  quantized CSI samples:  $T_s/2$  come from experiments collected in the empty room (E),  $T_s/4$  from those with a person sitting (S), and  $T_s/4$  from those with a person moving (M). This approach ensures the balance of the two situations we want to distinguish: empty (E) and occupied (O) room, which encompasses both S and M. One SVM is instantiated for each receiving antenna and the final classification is the result of majority voting.

Table X shows the per-group-of-experiments accuracy the SVM achieves on a training set with as many empty-room CSI's as our proposed method ( $M_T$  samples). However, since the SVM needs training samples from all situations to label, another  $M_T$  occupied-room CSI's are added to the set, coming 50/50 from the S and M situations for balance. Overall, the training set consists of  $2M_T$  CSI's. The proposed algorithm and the SVM achieve comparable performance across most experiment groups, resulting in a major improvement for group ID (e) and conversely in a much worse performance for group ID (g).

Considering the accuracy over the whole dataset (see Tab. XI), the two approaches achieve the same performance, but take two orders of magnitude less time for threshold tuning than SVM training. We have also tested the SVM with much smaller training sets ( $T_s = 200, 2000$ ). Interestingly, the performance does not decrease sharply decreasing the size of the training set; however, even limiting the system to  $T_s = 200$ , the training time remains over one order of magnitude larger than our method, making the proposed algorithm promising for embedded and IoT applications. A per-experiment-group breakdown of the SVM's training cost is discussed in Sect. VIII-C.

Table XI  
OVERALL ACCURACY, F1-SCORE, AND TUNING OR TRAINING TIME  
OBTAINED WITH THE PROPOSED ALGORITHM AND THE SVM TRAINED ON  
SETS WITH DIFFERENT NUMBERS OF CSI ( $T_s$ ).

|                             | <b>Our method</b><br>$T_s = M_T$ | <b>SVM</b><br>$T_s = 200$ | <b>SVM</b><br>$T_s = 2000$ | <b>SVM</b><br>$T_s = 2M_T$ |
|-----------------------------|----------------------------------|---------------------------|----------------------------|----------------------------|
| <b>Accuracy</b>             | 90.76%                           | 82.65%                    | 82.81%                     | 90.17%                     |
| <b>F1-Score</b>             | 92.92%                           | 87.11%                    | 87.48%                     | 93.30%                     |
| <b>Tuning/Training Time</b> | 0.025 s                          | 0.606 s                   | 1.014 s                    | 20.847 s                   |

This implementation is not the only way an SVM can

be used to sense an environment: for instance, [49] introduces kernels and uses Principal Component Analysis (PCA) to maintain low complexity in Human Activity Recognition (HAR) analysis. The use of SVM in [50], instead, comes after a data pre-processing through the symplectic transform, thus applying the ML method in the delay-Doppler domain and not directly to the collected CSI's.

For conventional supervised SVM-based sensing, such as the binary classifier considered in this work, the training set needs to include CSI samples from all output labels to allow to identify the separating hyperplanes. In real-world applications, this requires the end-user of the IoT sensing device to recreate all situations that need to be classified, collect enough data in each of them, and finally wait for the system training time. Although one-class SVMs can be trained using only samples from a nominal condition, they formulate the task as novelty detection rather than directly learning the distinction between empty and occupied environments, and their performance depends on whether all relevant occupied conditions produce sufficiently detectable deviations from the nominal CSI distribution. In contrast, the proposed algorithm leverages only few seconds of CSI's collected in the empty room, which is the simplest situation to recreate for any end-user and allows for inexpensive re-tuning any time the environment undergoes major modifications.

### B. Comparison with NN-based Classification

The natural benchmark for the proposed algorithm is provided by the SHARP activity recognition methodology [12]. The sophisticated feature extraction process, combined with a NN, achieves impressive classification accuracy, often exceeding 95%.

In contrast, the approach proposed in this work deliberately avoids complex learning pipelines. Despite its simplicity, the obtained results are competitive with those reported in [12], often reaching 100% accuracy. In particular, our distance-based metric enables human presence detection with consistent reliability across heterogeneous experimental conditions, encompassing time, space, and hardware variability. Although it does not aim for fine-grained activity recognition at a level comparable to a NN, it achieves comparable performance for coarser sensing tasks like occupancy detection.

For an accurate evaluation, Tab. IX can be compared with the results displayed in Tab. V of [12] obtained with the decision fusion strategy, which more closely resembles the majority voting approach implemented in this paper. The metric-based algorithm actually surpasses the NN's performance in both the bedroom (with LoS) and laboratory experimental setups, achieving 100% accuracy for the available experiments. Conversely, the bedroom setup without LoS makes situations harder to classify for both methods.

### C. Considerations on the Algorithm Complexity

The trade-off between accuracy and algorithm complexity is especially relevant when considering deployment constraints. SHARP is a time- and resource-consuming approach both in the training and inference phases. On the contrary, as

already noted, our distance-based metric does not require a full training the same way as a NN or an SVM require, since it only relies on a short threshold tuning phase.

In terms of computational complexity, normalization, quantization, and distance computation are all linear in the number of subcarriers. Therefore, the end-to-end cost of computing the distance between two CSI samples is  $\mathcal{O}(N_{sc})$ .

The complexity of the threshold tuning phase is instead a function of the number of distance values used for tuning,  $D_T$ . On each antenna, the threshold is computed from their mean and standard deviation, yielding linear time complexity  $\mathcal{O}(D_T)$ . The duration of this phase can be tailored to the specific deployment conditions by modifying  $D_T$  or by implementing an online threshold update mechanism as new CSI samples are collected. Finally, the complexity of the classification step is linear in the number of CSI samples. For an experiment yielding  $D_C$  distance values per antenna and  $N_A$  antennas, the classification has time complexity  $\mathcal{O}(N_A \cdot D_C)$ .

The first two columns of Tab. XII report the average processing time per CSI in each group of experiments<sup>4</sup>; it is obtained dividing each operation's execution time by the total number of CSI samples extracted for the considered setup, accounting for all receiving antennas. Compatibly with Matlab's numerical precision, the per-CSI processing times match the expected proportionality with the number of subcarriers imposed by the two considered 802.11 standards.

The two central columns of Tab. XII highlight the average per-CSI cost of computing the distance value and the overall duration of the threshold tuning phase for each group of experiments. The cost of distance computation is, as expected, proportional to  $N_{sc}$ . The cost of  $\mu_{th}$  tuning is instead dependent on the number of distance samples from each group of experiments, according to the number of available CSI samples: larger groups (e.g., ID: (g)) allow to use up to 50 000 samples for threshold tuning, while shorter experiments (e.g., ID: (c)) only allow to use  $\sim 6000$  samples. Once again, we stress that the threshold tuning phase can and should be adapted to the requirements of the target sensing application.

The time required by our proposed algorithm for threshold tuning can be directly compared with the SVM's training cost: the last column of Tab. XII shows that the SVM requires over two orders of magnitude longer to train than the  $\mu_{th}$  tuning.

An optimized algorithm implementation could further reduce the proposed method's computational requirements for IoT applications. Since the discussed algorithm is merely an example of how the stochastic characterization of the CSI can inform sensing tools, further code optimization for real implementations falls outside of the scope of this study.

## IX. CONCLUSIONS

This work contributes to the understanding and interpretation of the potential of CSI-based sensing by introducing a unified stochastic characterization of Wi-Fi CSI subcarrier amplitude increments. So far, AI-based environment sensing

Table XII  
AVERAGE PER-CSI EXECUTION TIME OF NORMALIZATION (NORM), QUANTIZATION (QUANT), AND DISTANCE COMPUTATION (DIST). THRESHOLD TUNING AND SVM TRAINING TIME ARE MEASURED ON  $T_s = M_T$  AND  $T_s = 2M_T$ , RESPECTIVELY. EXPERIMENT GROUP IDS ARE THE SAME AS TABS. IX AND X.

| Shared preprocessing |                  |                   | Our method       |                   | SVM           |
|----------------------|------------------|-------------------|------------------|-------------------|---------------|
| ID                   | Norm ( $\mu s$ ) | Quant ( $\mu s$ ) | Dist ( $\mu s$ ) | $\mu_{th}$ Tuning | Training Time |
| (a)                  | 0.121            | 0.051             | 0.268            | 1.129 ms          | 346 ms        |
| (b)                  | 0.243            | 0.100             | 0.672            | 0.846 ms          | 226 ms        |
| (c)                  | 0.500            | 0.204             | 1.438            | 0.665 ms          | 227 ms        |
| (d)                  | 0.437            | 0.136             | 1.918            | 3.974 ms          | 915 ms        |
| (e)                  | 0.414            | 0.081             | 2.240            | 2.027 ms          | 615 ms        |
| (f)                  | 0.485            | 0.203             | 1.884            | 1.822 ms          | 522 ms        |
| (g)                  | 0.142            | 0.020             | 0.285            | 4.206 ms          | 12 800 ms     |
| (h)                  | 0.142            | 0.021             | 0.288            | 1.776 ms          | 2620 ms       |
| (i)                  | 0.151            | 0.024             | 0.317            | 0.889 ms          | 1356 ms       |
| (j)                  | 0.141            | 0.020             | 0.291            | 3.167 ms          | 1228 ms       |

techniques have delivered impressive results across multiple tasks, including environment recognition, positioning, and movement detection, to mention a few. However, the limited knowledge on the internal reasoning behind the outcomes produced by these systems often hinders the interpretability of the results. This lack of transparency makes it difficult to discern what features of the CSI samples catch the attention of AI tools and support their decisions. Moreover, the lack of literature on the long-term stochastic characterization of propagation environments limits the interpretations of sensing results.

This work has shown that the CSI carries a huge amount of information, making it difficult to proceed with direct numerical evaluations of the probability that a CSI was collected in a specific environment. The key result is that CSI amplitude evolution can be described through its short-term variability, the increments. In particular, the increments distribution exhibits heavy tails that are accurately modeled by a T Location Scale distribution across all considered experimental setups, regardless of differences in bandwidth, hardware, inter-frame time, and environmental dynamics. The analysis of the amplitude autocorrelation reveals the presence of residual memory that persists over large lags and is inconsistent with exponential decay, supporting a non-Markovian interpretation of the CSI amplitude evolution. Consistently with this result, the increments autocorrelation exhibits a significant one-step negative memory, which is a direct consequence of the non-exponential decay of  $\rho_A$ .

This stochastic characterization enables a compact representation of the CSI's through the quantization of their amplitude, supporting the definition of a simple distance metric between samples captured in different scenarios. The simple sensing algorithm proposed serves as an example of how the statistical characterization can inform the design of lightweight, interpretable tools to complement AI-based sensing methods: its resource efficiency makes it suitable for deployment in IoT systems, where energy efficiency and algorithm simplicity are critical. By reducing the dependency on complex, time and energy-consuming AI models, this method enables transparent and resource-aware implementations of movement detection

<sup>4</sup>All code ran on an Apple MacBook Pro with Apple M4 Pro chip and 48GB of RAM. Matlab version is R2025a. Execution times are measured using Matlab's built-in Profiler tool.

techniques in contexts that impose strict resource constraints. Overall, this approach contributes to the development of IoT technologies and opens to the exploration of JCAS for everyday applications.

Finally, the extensive, public CSI repository that we publish associated to this paper contains all the experimental data collected for this work, encompasses different well-documented scenarios, allows easy reproduction of the results, and further research on this very interesting topic.

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## APPENDIX A DERIVATION OF EQ. (16)

We hereby provide a full expansion of the calculations that lead to Eq. (16). Define

$$\begin{aligned}
 \mu_{\delta(\cdot, n)} &= \mathbb{E}[\delta(\cdot, n)] \\
 &= \mathbb{E}[A(k, n) - A(k-1, n)] \\
 &= \mathbb{E}[A(k, n)] - \mathbb{E}[A(k-1, n)] \\
 &= \mu_{A(\cdot, n)} - \mu_{A(\cdot, n)} = 0.
 \end{aligned}$$

Then, represent the auto-covariance of the increments as a function of the auto-covariance of the amplitude of the CSI

on subcarrier  $n \in \left[-\frac{N_{sc}}{2}, \frac{N_{sc}}{2} - 1\right]$ :

$$\begin{aligned}
 R_{\delta}(\tau, n) &= \mathbb{E}[\delta(k, n) \cdot \delta(k - \tau, n)] - \mu_{\delta(\cdot, n)}^2 \\
 &= \mathbb{E}[\delta(k, n) \cdot \delta(k - \tau, n)] \\
 &= \mathbb{E}[(A(k, n) - A(k-1, n)) \\
 &\quad \cdot (A(k - \tau, n) - A(k - 1 - \tau, n))] \\
 &= \mathbb{E}[A(k, n) \cdot A(k - \tau, n) \\
 &\quad - A(k, n) \cdot A(k - 1 - \tau, n) \\
 &\quad - A(k - 1, n) \cdot A(k - \tau, n) \\
 &\quad + A(k - 1, n) \cdot A(k - 1 - \tau, n)] \\
 &= \mathbb{E}[A(k, n) \cdot A(k - \tau, n)] \\
 &\quad - \mathbb{E}[A(k, n) \cdot A(k - 1 - \tau, n)] \\
 &\quad - \mathbb{E}[A(k - 1, n) \cdot A(k - \tau, n)] \\
 &\quad + \mathbb{E}[A(k - 1, n) \cdot A(k - 1 - \tau, n)] \\
 &= \mathbb{E}[A(k, n) \cdot A(k - \tau, n)] \\
 &\quad - \mathbb{E}[A(k, n) \cdot A(k - (1 + \tau), n)] \\
 &\quad - \mathbb{E}[A(k - 1, n) \cdot A(k - 1 - (\tau - 1), n)] \\
 &\quad + \mathbb{E}[A(k - 1, n) \cdot A(k - 1 - \tau, n)]
 \end{aligned}$$

Recalling and inverting the definition of auto-covariance, we have  $\mathbb{E}[A(k, n) \cdot A(k - \tau, n)] = R_A(\tau, n) + \mu_{A(\cdot, n)}^2$ . Plugging this in the equations above, we obtain:

$$\begin{aligned}
 R_{\delta}(\tau, n) &= R_A(\tau, n) + \mu_{A(\cdot, n)}^2 \\
 &\quad - R_A(\tau + 1, n) - \mu_{A(\cdot, n)}^2 \\
 &\quad - R_A(\tau - 1, n) - \mu_{A(\cdot, n)}^2 \\
 &\quad + R_A(\tau, n) + \mu_{A(\cdot, n)}^2 \\
 &= 2 \cdot R_A(\tau, n) - R_A(\tau + 1, n) \\
 &\quad - R_A(\tau - 1, n).
 \end{aligned}$$

Computing this for  $\tau = 0, 1$ , we have:

$$\begin{aligned}
 R_{\delta}(0, n) &= 2 \cdot R_A(0, n) - R_A(1, n) - R_A(-1, n) \\
 &= 2 \cdot R_A(0, n) - 2 \cdot R_A(1, n), \\
 R_{\delta}(1, n) &= 2 \cdot R_A(1, n) - R_A(2, n) - R_A(0, n).
 \end{aligned}$$

To compute the autocorrelation, by definition we divide the auto-covariance at lag  $\tau$  by the auto-covariance at lag 0:

$$\begin{aligned}
 \rho_{\delta}(0, n) &= \frac{2 \cdot R_A(0, n) - 2 \cdot R_A(1, n)}{2 \cdot R_A(0, n) - 2 \cdot R_A(1, n)} = 1 \\
 \rho_{\delta}(1, n) &= \frac{2 \cdot R_A(1, n) - R_A(2, n) - R_A(0, n)}{2 \cdot R_A(0, n) - 2 \cdot R_A(1, n)}.
 \end{aligned}$$

Knowing that Eq. (3) can be inverted to

$$R_A(\tau, n) = \rho_A(\tau, n) \cdot R_A(0, n),$$

we obtain:

$$\begin{aligned}
 \rho_{\delta}(1, n) &= \frac{R_A(0, n) \cdot (2\rho_A(1, n) - \rho_A(2, n) - 1)}{2R_A(0, n) \cdot (1 - \rho_A(1, n))} \\
 &= \frac{2\rho_A(1, n) - \rho_A(2, n) - 1}{2 - 2\rho_A(1, n)}.
 \end{aligned}$$

As discussed in Sect. III, the denominator of this fraction is always positive, assuming  $\rho_A(1, n) < 1$ , while the dominating

term of the numerator is  $\rho_A(0, n) = -1$ , resulting in an overall negative fraction that is coherent with the observed phenomenon.

This interesting dependence of the increments autocorrelation on the amplitudes autocorrelation, which, to the best of our knowledge, is highlighted here for the first time in the analysis of CSI behavior, has clear consequences on sensing functions based on CSI analysis.



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